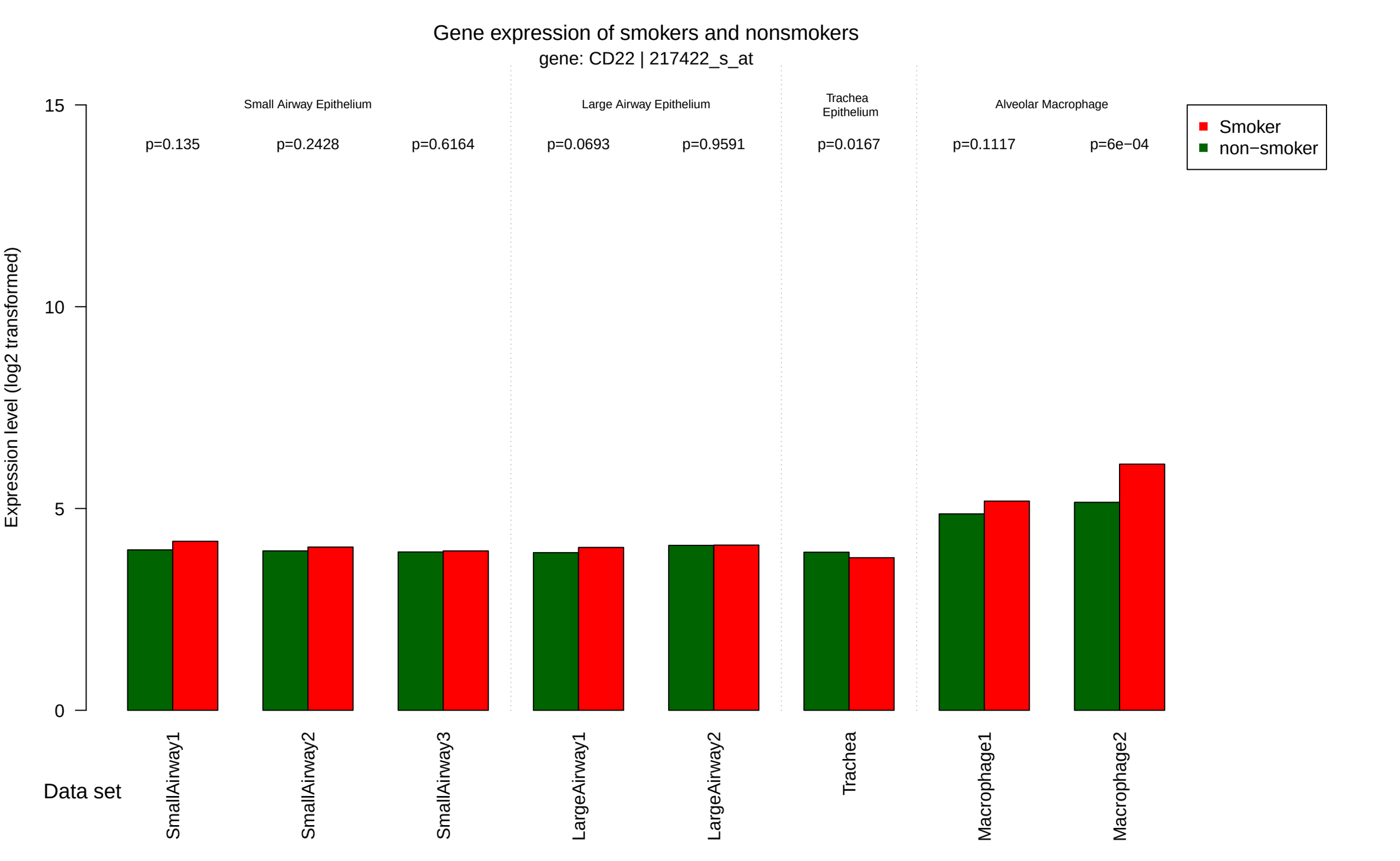
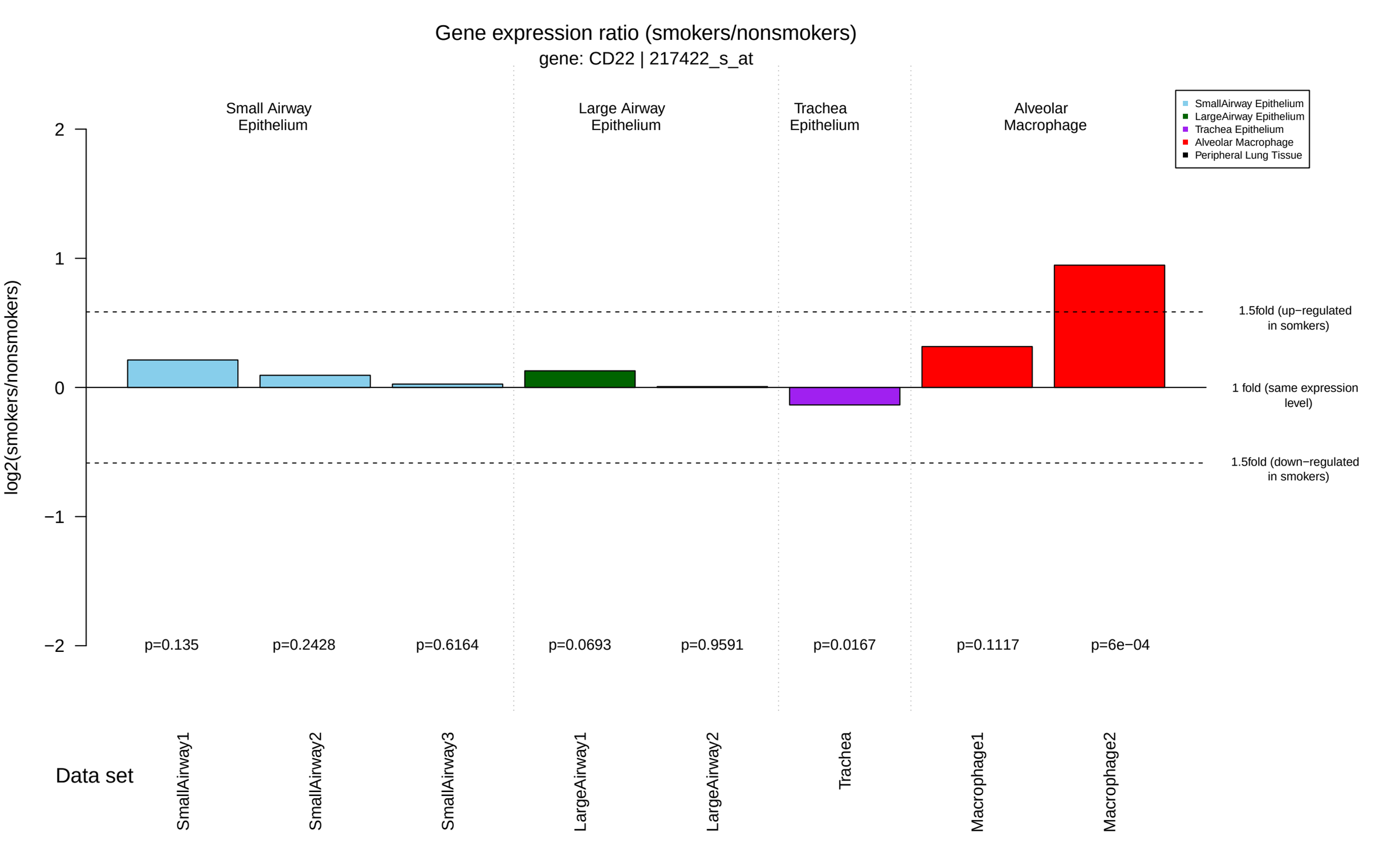
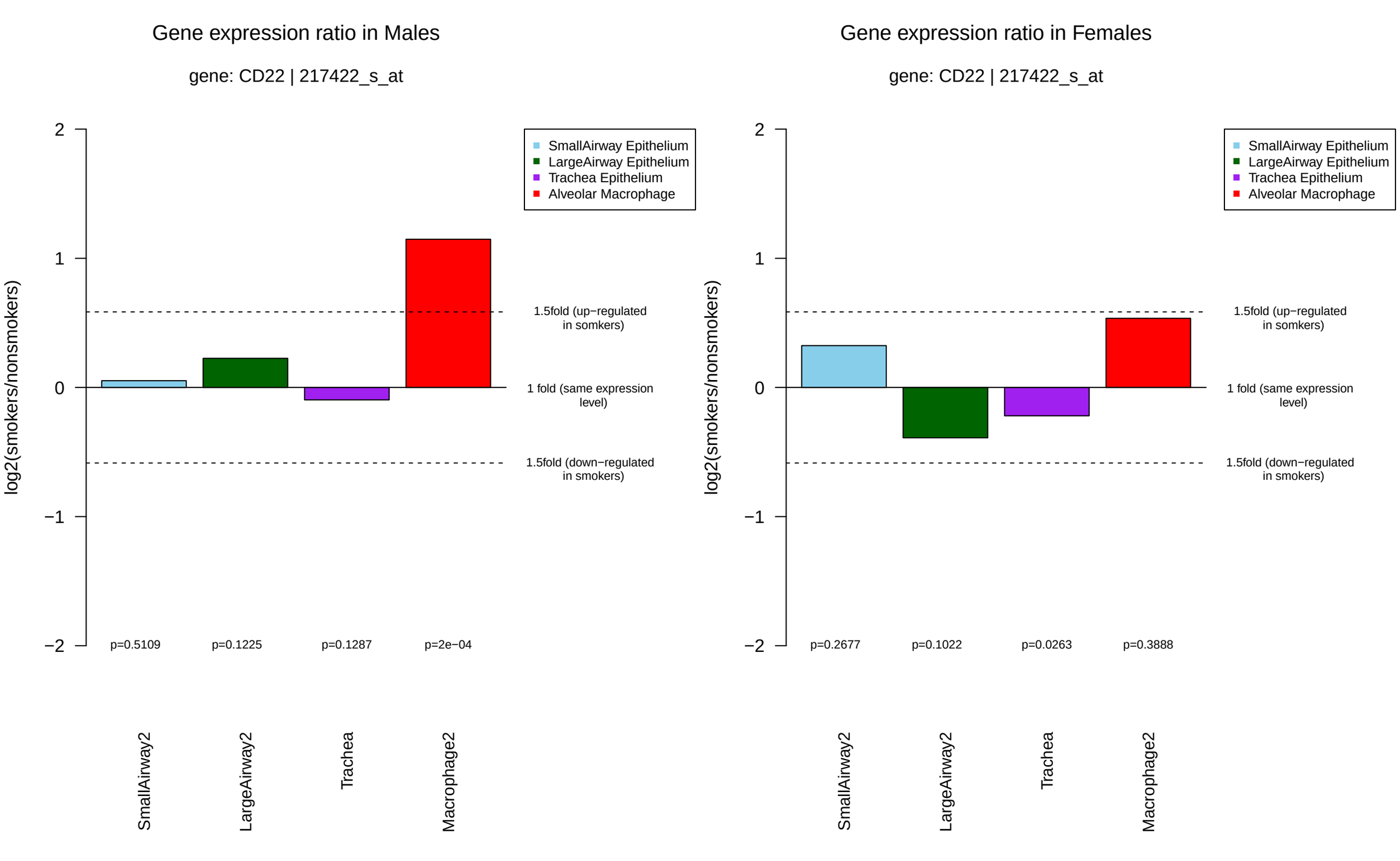
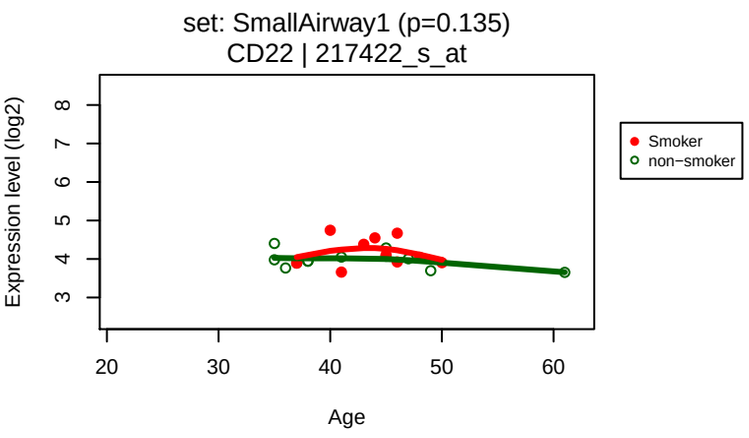
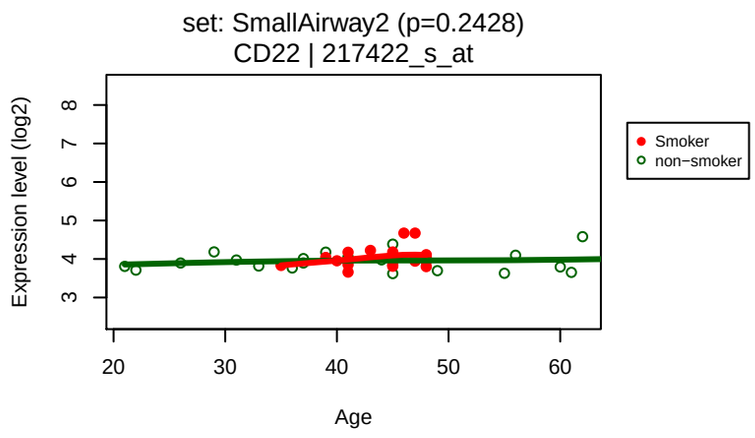
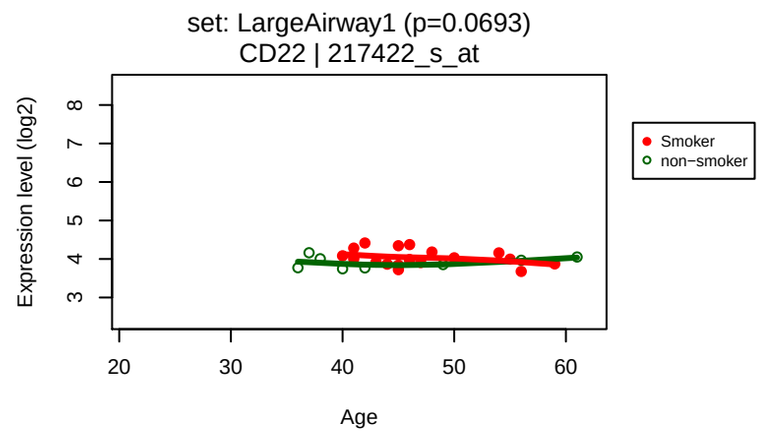

Gene expression of smokers and nonsmokers
gene: CD22 | 217422_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.135
p=0.2428
p=0.6164
p=0.0693
p=0.9591
p=0.0167
p=0.1117
p=6e−04
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

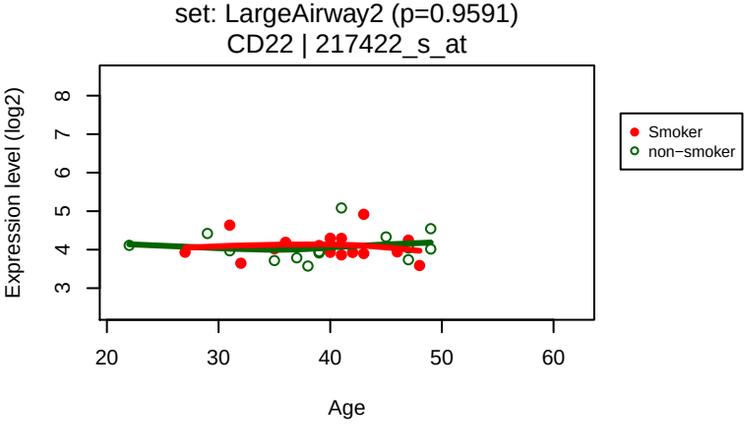

Gene expression ratio (smokers/nonsmokers)
gene: CD22 | 217422_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.135
p=0.2428
p=0.6164
p=0.0693
p=0.9591
p=0.0167
p=0.1117
p=6e−04
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

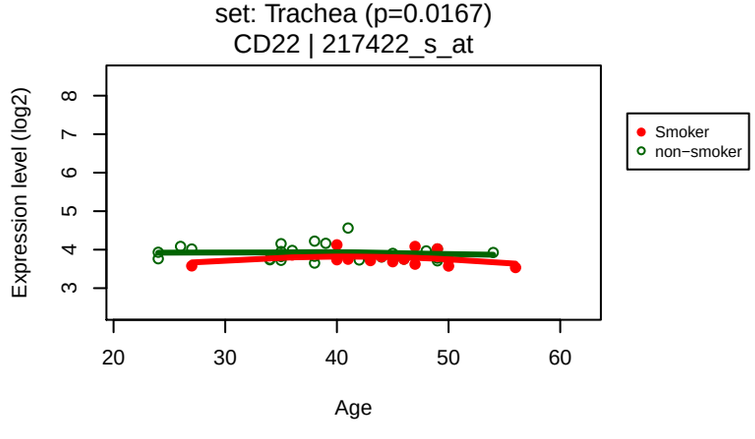

Gene expression ratio in Males
gene: CD22 | 217422_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5109
p=0.1225
p=0.1287
p=2e−04
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CD22 | 217422_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2677
p=0.1022
p=0.0263
p=0.3888
SmallAirway2
LargeAirway2
Trachea
Macrophage2

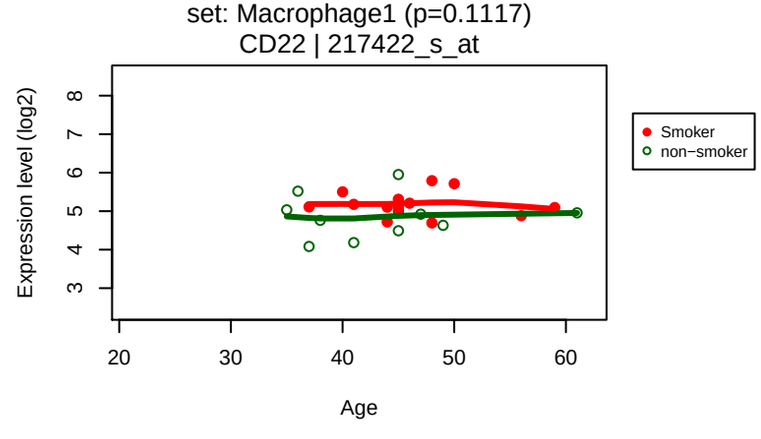

set: SmallAirway1 (p=0.135)
CD22 | 217422_s_at
Expression level (log2)
Age
Smoker
non−smoker

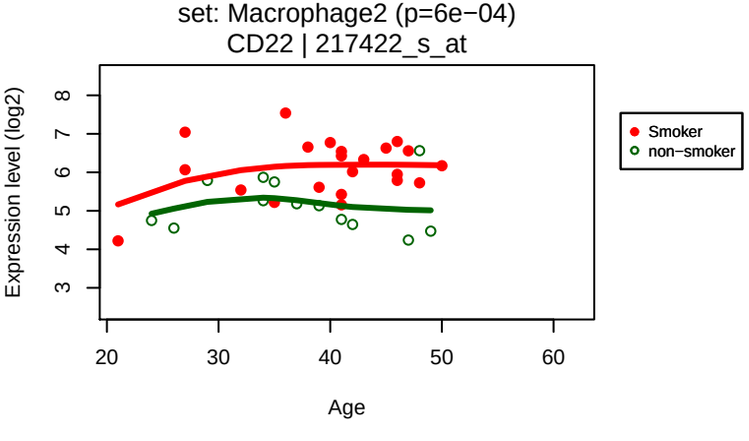

set: SmallAirway2 (p=0.2428)
CD22 | 217422_s_at
Expression level (log2)
Age
Smoker
non−smoker

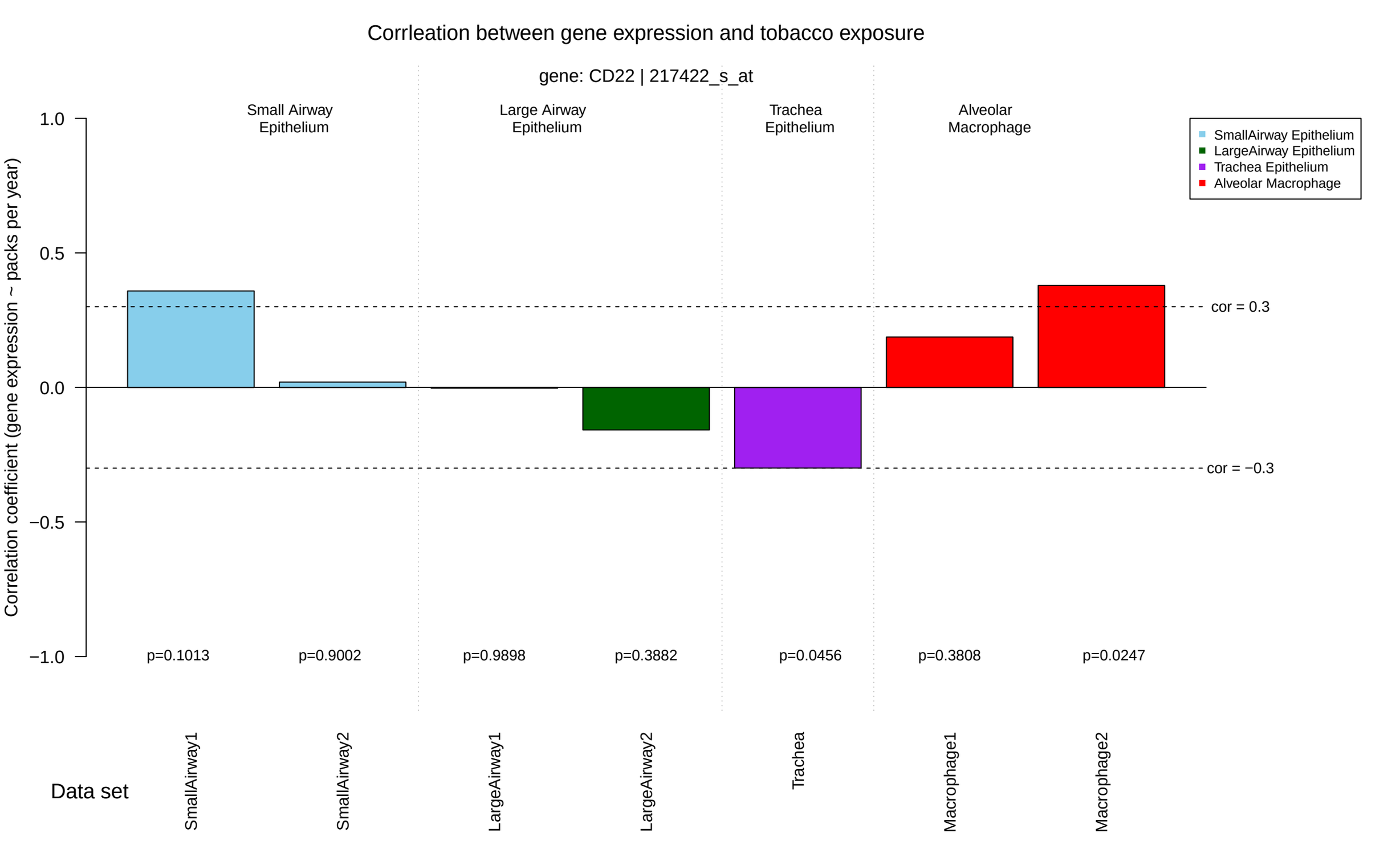

set: LargeAirway1 (p=0.0693)
CD22 | 217422_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9591)
CD22 | 217422_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0167)
CD22 | 217422_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1117)
CD22 | 217422_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=6e−04)
CD22 | 217422_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CD22 | 217422_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1013
p=0.9002
p=0.9898
p=0.3882
p=0.0456
p=0.3808
p=0.0247
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2